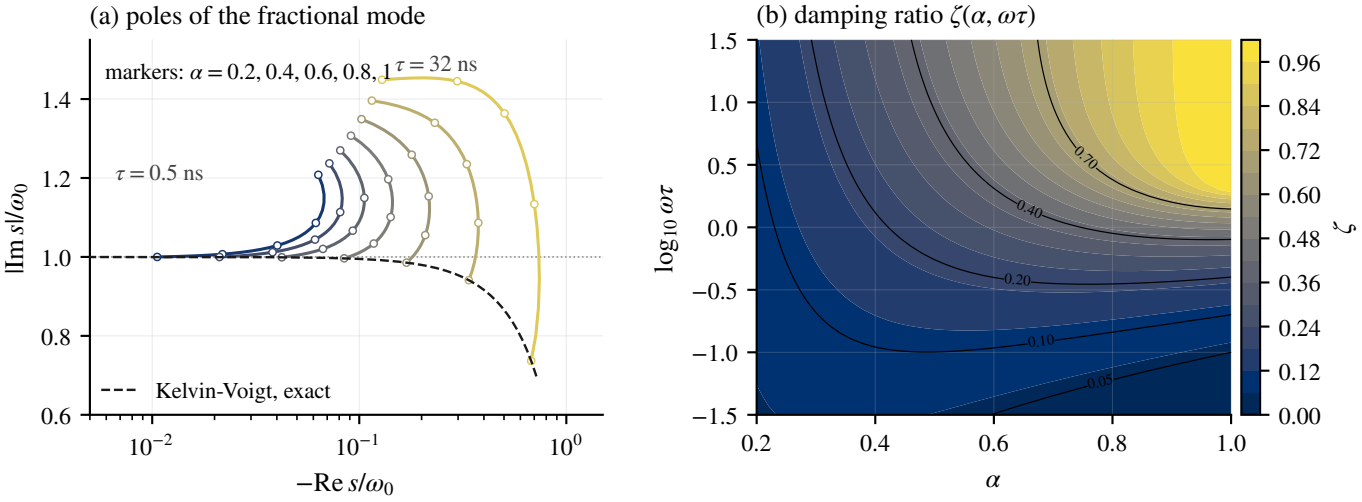
(a) poles of the fractional mode
markers: $\alpha = 0.2, 0.4, 0.6, 0.8, 1$
$\tau = 32$ ns
$\tau = 0.5$ ns
Kelvin-Voigt, exact
$|\mathrm{Im}\, s|/\omega_0$
$-\mathrm{Re}\, s/\omega_0$
1.4
1.2
1.0
0.8
0.6
$10^{-2}$
$10^{-1}$
$10^{0}$
(b) damping ratio $\zeta(\alpha, \omega\tau)$
$\log_{10} \omega\tau$
$\alpha$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0.2
0.4
0.6
0.8
1.0
0.70
0.40
0.20
0.10
0.05
$\zeta$
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00